



RN00092

Release Notes for S32R41 Radar SDK

Rev. 0.8.10 – December 18th 2024

Release Notes

Document information

Info	Content
Keywords	Radar SDK, S32R41
Abstract	Release notes describing release contents, changes, limitations and known issues.



Table of Contents

Table of Contents2

1. Read Me First.....3

 1.1 Environment Requirements3

 1.2 Build configuration.....3

 1.3 Support information.....4

2. Release Description5

 2.1 List of Modules5

 2.2 New Features6

 2.3 Fixed Issues7

 2.4 Known Issues and Limitations7

 2.5 Complete List of Supported Features8

3. Upgrading from the Previous Release15

 3.1 General Guidelines15

4. Usage of Third Party Software and Licensing16

Appendix A: Document History17

Appendix B: Package Release History17

5. Legal information.....19

 5.1 Definitions19

 5.2 Disclaimers.....19

1. Read Me First

This release of the **Radar SDK** software package for S32R41 hardware platform provides updated software enablement for the SPT, BBE32 signal processing accelerators and TEF82XX radar transceiver.

It can be used by any party interested in accessing radar transceiver and data processing functionalities from the S32R41 ARM M7 host processor in a NXP RTOS environment.

If you are new to RSDK, please start by getting familiarized with the [RSDK User Manual](#) (html format) and example applications (located in [/Apps/](#)).

1.1 Environment Requirements

The following dependencies are used for building and executing the binaries in the software package:

Software:

- **NXP S32 Design Studio Version: 3.5.13** (updated from 0.8.9); Build id: 240726, including:
 - S32R41 development package; Version: 3.5.3; Build id: 202306291744
 - Radar extension package for S32R41; Version: 1.1.0 RTM; Build id: 202306291243
 - NXP GCC for Arm Embedded Processors v9.2 build 1649; Version: 1649; Build id: 202307190917
 - BBE32 DSP Add-On Package for S32R41; Version: 1.1.0 RTM; Build id: 202306291245
 - **S32 Design Studio Platform Tools package; Version: 3.5.13** (new from 0.8.9); Build id: 202407261948
- EB Tresos 29.0.0
- **NXP Real-time Drivers for S32R AUTOSAR R21-11 Version P03; Version: 2.0.0**(updated from 0.8.9); Build id: 202404120938
- **NXP RTOS for S32R41 Version: 1.0.0**(updated from 0.8.9); Build id: 202402291554
- **SW32R41 IPCF 1.1.0 D2405** (updated from 0.8.9);
- Lauterbach Trace32 PowerView for ARM, Software Version: N.2022.08.000150923, or newer
- (optional) Matlab R2021a 64bit
- (optional) Matlab Signal Processing Toolbox v8.6 and Model-Based Design SPT Toolbox v1.7.0, used by the Matlab bit-exact model

Hardware:

- Host PC running Windows 10
- S32R418AA processor E5, mounted on S32R41 EVB board Rev B or newer
- TEF82XX radar front-end IC version ES2.2 mounted on Customer Antenna Board (CAB) + S32R41 TEF82XX adapter board
- Lauterbach Power Debug Interface and cable (with license for A53 and BBE32 debugging) or NXP - DS Debugger

1.2 Build configuration

This release was developed and tested with:

- NXP GCC for Arm Embedded Processors v9.2 build 1649; Version: 1649; Build id: 202307190917

The build options are available in the S32 Design Studio project files of each software component (.cproject), or alternately in the makefiles, where available.

1.3 Support information

Please contact your friendly sales representative or access <https://www.nxp.com/> support page. When reporting problems (we thank you for that!), please include:

- RSDK release ID
- tools used for rebuilding the RSDK artefacts
- target hardware platform (e.g. platform, board revision)
- any dependencies or modifications you may have added or applied to create your environment

2. Release Description

2.1 List of Modules

CATEGORY	DELIVERABLE	LOCATION	STATUS VS PREV RELEASE	QA LEVEL
Example Apps	Front-end data capturing and processing example	/Apps/RFE_processing_example	Updated	reference code
	Data processing example without a connected front-end	/Apps/RSDK_offline_example	Updated	reference code
	Example app for creating a BBE32 executable image	/Apps/DSP_Dispatcher_example	Updated	reference code
Docs	User manual – html format	/Docs/index.html or /Docs/RSDK_User_Manual/index.html	Updated	--
Software Components	SPT driver	/SPT/SPT_driver	Unchanged	PRC
	SPT reference kernels	/SPT/SPT_kernels	Updated	reference code
	DSP Dispatcher	/DSP/DSP_dispatcher	Updated	PRC
	DSP Host Driver	/DSP/DSP_host_driver	Updated	PRC
	DSP Radar Baseband Algorithm library	/DSP/DSP_radar_bb_algos	Updated	reference code
	DSP Linear Algebra Library	/DSP/DSP_la	Updated	PRC
	RFE Abstract 2.0 Driver	/RFE_abstract/RFE_driver2	Updated	PRC
	CSI2 Driver	/CSI2/CSI2_driver	Updated	PRC
	CTE Driver	/CTE/CTE_driver	Updated	PRC
Dev Tools	RSDK Trace logger and parser	/Tools/Trace /Tools/Trace_parser	Unchanged	reference code
	C Model for DSP Linear Algebra Library	/Tools/C_model	Unchanged	reference code
	SPT bit-exact matlab kernel models	/Tools/SPT_bitexact_model	Unchanged	reference code
	Matlab utility functions	/Tools/Matlab_aux	Unchanged	reference code
	RFE Abstract 2.0 Config Generator	/Tools/RFE2_Config	Updated	reference code

The following sections provide a detailed description of release content and incremental changes.

2.2 New Features

This release provides updates for the following components:

CSI2

Code updates for simple bug fixing.

CTE

Code updates for simple bug fixing.

RFE Abstract 2.0

- General module updates :
 - Added support for TX Calibration with reduced output power transmission.
 - Improved compliancy to the Software Safety manual by adding support for Toggletest between Leader (MCU_INT) and Follower (CHIRP_START_IN) for cascaded systems.
 - added a second option for data transfer/communication between API and FW, NXP – IPCF (InterProcessor Communication Framework)
 - Code updates for MISRA compliance
 - Added ATB voltage measurements to GetRegisterDump function
 - Added CRC verification at blob generation/usage
 - Timing optimizations for RFE FSM
 - RXBist center frequency can be changed via XML. RSDK-11130
 - Improved phase consistency between chirp sequences. RSDK-11382
- API updates :
 - Front-end registers dumping (CDD_Rfe_GetRegisterDump) function name update (from CDD_Rfe_GetRegistryDump)
 - RFE interrupt on firmware events and API command completion (CDD_Rfe_ConfigureInterrupts)
 - Fusa fault statistics for R1 faults (CDD_Rfe_FuSaFaultStatistics). RSDK-10901

SPT kernels

- Extended 3DFFT to double the sample size.

BBE32 DSP

- **DSP Dispatcher**
 - Added support to provide information when DSP enters IDLE state.
 - Added support to initialize prefetch aggression.
- **DSP Host Driver**
 - Added support to provide information when DSP enters IDLE state.
- **DSP Radar Algos Lib**
 - Better organization of the library based on platform
 - FFT2real kernels return the lower half of the two resulting FFTs.
- **DSP Linear Algebra Lib**
 - Performance optimizations on kernels

Tools – RFE Abstract 2.0 Config Generator

For a detailed list of changes, please look at the Config Generator User Manual located in */Tools/RFE2_Config* folder

- Several Parameters added:
 - txCalibration – allows calibration with normal/reduced TX transmission
 - txCalibrationThresholds – configures thresholds for *txCalibration with reduced transmission*, that trigger automatic TX power adjustment during calibration
- *pll-loop-filter-bandwidth* – value limit removed, but outputs warning if using settings that yield higher phase noise
- Tool outputs a warning if using different center frequencies in the same sequence
- *frequency-drift-steps* – range updated. Values now represent an increment of 42.9153 Hz.
- Tool outputs an error if *chirp-interval-time-ticks* is smaller than the sum of its parts
- Tool outputs an error if setting *dwel-time-ticks* & *settle-time-ticks* & *reset-time-ticks* < 160 (TEF82xx internal limit)
- Parameters under <rxBist> element, have been changed to integers, and the values represent degrees (for phase). See the manual mentioned above this list of changes, for details and examples.
- Reduced blob size by 33 bytes
- Radar Cycle Schedule Generator will output a warning if a minimum idle time of 50µs is not ensured, for each sequence
- Added CRC to the blob, to ensure data consistency between the M7 cores.
- Added details about the structure of the Dynamic Table data, in the Config Generator User Manual

Example apps

- Added RFE register dump in case of FuSa in RFE example

2.3 Fixed Issues

ID	Module	Description
RSDK-10293	RFE Abstract 2.0	Call CDD_Rfe_GetBistZeroHourReferenceData return E_NOT_OK when run with RFE FW single mode
RSDK-9365	RFE Abstract 2.0	Build fails RFE-FW with GHS compilers
RSDK-10861	RFE Abstract 2.0	Chirp start line Toggle Test between leader and follower devices is not enabled by default.
RSDK-10873	RFE Abstract 2.0 Config Generator	CDD_Rfe_UpdateParam() can set Dwell, Settle and Reset time to a lower value than what should be allowed, because the Min/Max blob (that contains range check values) is not set correctly for those parameters.
RSDK-10981	DSP dispatcher	DspExceptionHandler does not save exception context correctly
RSDK-10666	RFE Abstract 2.0	FuSa faults that were not contained in the XML, were automatically masked

2.4 Known Issues and Limitations

ID	Affected Modules	Description
RSDK-10015	RFE Abstract 2.0	Using different center frequencies per chirp sequence causes unexpected spurs in CW mode
RSDK-11478	SPT	SPT Write issue with Non-blocking DSP Commands

RSDK-11847	RFE Abstract 2.0	Config Generator outputs warning about using Frequency Drift with different Center Frequencies, even if the profiles are not in the same sequence.
RSDK-11729	RFE Abstract 2.0	Add API for updating dynamic table parameters (TEF82xx IQ phase control) at runtime.
RSDK-11881	RFE Abstract 2.0	Incorrect rxBistPhaseThreshold value conversion in BistUpdate_1 function

2.5 Complete List of Supported Features

Module	Features	Introduced in release
SPT driver	Provide an interface to access the SPT functionalities (implemented via SPT kernels), exposed on host ARM processor. Single core usage (A53). Zephyr OS support API option to boot the BBE32 DSP and trigger remote execution of algorithms. Blocking and non-blocking operating modes. API command to enable memory parity error injection API command to trigger an event on the SPT software event line (SW_EVTREG register). Supporting all interrupts triggered by the SPT hardware module. Enabled all error interrupts except 'memory lock' errors. API Callback functions provided for the SPT interrupt handlers. Control of API call sequence in a multithreaded OS environment. Guard each Driver API call with a check for SPT hw reset state Detection mechanism for spurious STOP interrupts BBE32 Errors are sent to user app through SPT Driver DSP interrupt	0.8.0
	Internal code refactoring	0.8.1
	Migrated SPT driver to AUTOSAR: SPT Complex Device Driver and Tresos plugin support	0.8.2

Module	Features	Introduced in release
	Updated Complex Device Driver and plugin for AUTOSAR R21-11 Improved error reporting mechanism – added more error codes	0.8.4
SPT kernels library	The <i>SPT kernels</i> perform Range FFT, Doppler FFT, Antenna Combining (Non Coherent or FFT-based), Peak Search, Digital BeamForming and Direction of Arrival for the following use-cases: • 512 samples/chirp, 128 chirps, 4 physical channels • 256 samples/chirp, 256 chirps, 4 physical channels • 1024 samples/chirp, 128 chirps, 4 physical channels • 256 samples/chirp, 256 chirps, 4 physical channels, 2TX TD-MIMO • 256 samples/chirp, 128 chirps, 4 physical channels, 3TX TD-MIMO • 512 samples/chirp, 128 chirps, 8 physical channels • 256 samples/chirp, 256 chirps, 8 physical channels • 1024 samples/chirp, 128 chirps, 8 physical channels • 256 samples/chirp, 256 chirps, 8 physical channels, 2TX TD-MIMO • 256 samples/chirp, 128 chirps, 8 physical channels, 6TX TD-MIMO • 1024 samples/chirp, 256 chirps, 4 physical channels, 3TX DD-MIMO • RF Build-in-self-test, 2048 samples, 4 channels	0.8.0
	• RF Built-in-self-test, 128 samples, 4 channels	0.8.0 CD2
	Added OS-CFAR kernel, which combines SPT and BBE32 processing	0.8.1
	New Radar uses-case supported: 1024samples/chirp, 64 chirps, 4 physical channels, 4TX TD-MIMO	0.9.0
	Added Matlab bit-exact equivalents. Available for: • 512 samples, 128 chirps, 4 channels • 256 samples, 256 chirps, 4 channels • 1024 samples, 128 chirps, 4 channels • 512 samples, 128 chirps, 8 channels • 256 samples, 256 chirps, 8 channels • 1024 samples, 128 chirps, 8 channels	0.9.0
	Added support for CSI2 statistics in Range FFT SPT kernels.	0.8.2
	New kernels: - o Init, Range, Doppler, NCC for 1024 samples, 64 chirps, 4 channels, CP4Q - o Init, Range, Doppler, NCC, PS for 2048 samples, 256 chirps, 4 channels, CP4Q	0.8.4
	Extended 3DFFT to double the sample size.	0.8.10
DSP Dispatcher	BBE32 DSP is integrated with SPT and A53 core. DSP calling convention allows parameter passing directly from SPT kernels, or indirectly from A53 (SPT Driver). DSP Dispatcher API allows enablement of SPT Event interrupts and callbacks. RSDK BBE32 projects available in NXP Design Studio. DSP image file for A53 executable is generated in fully automatic mode. Built with safety-certified build toolchain (LLVM-based) Safety-related features: - user-defined BBE32 MPU configuration. - interrupt configuration and custom exception handlers in the DSP Dispatcher - SPT-DSP Calling convention includes an error detection mechanism	0.8.0
	Added new task for running periodic jobs, scheduled by CSI2 radar frame (VSYNC) interrupt Added handler for interrupts triggered by the A53 core and CSI2 radar frame sync (VSYNC).	0.8.0 CD2

Module	Features	Introduced in release
	Removed code that disables/re-enables the SPT command queue interrupt, because it's cleared automatically by the XOS low-level handler.	0.8.1
	Expanded multitasking support: DSP exceptions are now reported to the CPU Send ACK after DSP asynchronous commands are executed	0.9.0
	Added P1 (chirp end) and P5 (long jobs) tasks Added P1 preempt/non-preempt options Added detection mechanism for DSP arithmetic exceptions and rounding configuration Added configuration command that controls DSP behavior Added ACK for every DSP Dispatcher command Added CRC check for every command Fixed a bug that didn't allow batch updating multiple job lists (all jobs should have their own application buffer) IPCF integration for communication with DSP Host Driver	1.0.0
	Switched ASYNC_DONE message to another IPCF channel DSP communication is now handled by IPCF managed channels (protection for multiple burst messages) DSP error messages are now correctly handled when burst messages occur	0.8.2
	Fixed MPU set-up. Runtime MPU was not working due to Xtensa limitations. Updated IPCF configuration to support new interrupt registration convention.	0.8.4
	Added support to provide information when DSP enters IDLE state. Added support to initialize prefetch aggression.	0.8.10
DSP Radar Baseband Algorithms	Added new functions for CA-CFAR and OS-CFAR	0.8.1
	Added new functions for FFT128/256/512/1024 24b complex Async tasks from CPU -> BBE32 are now correctly handled inside P3 task. Added exception reporting from BBE32 -> CPU. Added ACK for P3 (ASYNC) task.	0.9.0
	Added new function for FFT2048 24b complex	1.0.0
	Added 24bit real FFTs. Supported lengths: 128/256/512/1024/2048.	0.8.2
	FFT routines now using DRAM for intermediate (between rounds) data storing (scratch memory)	0.8.3
	Optimized FFTs Added wrapper on top of FFTs routines Added 24bit real Histogram kernel and wrapper	0.8.4
	Added DoA kernels and wrappers for 128/256 samples (FFT based implementation). Optimized real FFT kernels + split on multiple functions. Added windowing support for FFTs. All wrappers for complex FFTs were combined in 1 wrapper. All wrappers for real FFTs were combined in 1 wrapper. RsdkBbe32FftInit wrapper added for moving twiddles and window coefficients from SRAM to DRAM.	0.8.9

Module	Features	Introduced in release
	The library has been reorganized based on platform. All FFTs kernels and wrappers have been reorganized into two kernels and two wrappers for complex and real input cases. (supporting all previous sizes). FFT2real kernels return the lower half of the two resulting FFTs.	0.8.10
BBE32 DSP Linear Algebra (LAL)	The DSP library for Linear Algebra functions on parallel architecture SIMD. The functions are using the Vector Floating Point Unit from the BBE32EP DSP. The VFPU is processing 8 parallel streams (data) in one cycle. The functions in the DSP LAL library are the following: - Trigonometric functions approximated by Taylor series (sin, cos, arcsine, arctangent, arctangent2); - Logarithm functions approximated by Taylor series (base 2,10,natural); - Exponential function approximated by Taylor series (real) - Root square functions (real or complex); - Elementary functions (add, subtraction, reciprocal, division, modulo, min, max – real or complex); - Matrix and vector multiplications with conjugate options (real or complex); - Reduction functions: product, sum, norm, sum of squares, sum of cubes (real or complex); - Cross correlation (real or complex) and cross-conjugate correlation; - Cross product (real or complex) and cross-conjugate product; - Sort functions – Odd-Even sort (ascending/descending); - Bitmask threshold functions; - Data transformation functions (copy, fill, identity matrix, get/set cell, deinterleaving, interleaving, get imaginary/real part of complex number); - Matrix decompositions: QR, Cholesky (real or complex); - Matrix inversion: for triangular matrices and for non-singular matrices using QR decomposition or Gauss elimination (real or complex); - Determinant of a square matrix by QR decomposition (real or complex); - Eigensolvers: for symmetric/hermitian matrices and for complex upper Hessenberg matrices; - Statistical functions: average (arithmetic, weighted, harmonic, root mean square – real or complex). - FFT stage 2 twiddle factor generator; - Spatial interpolation (complex).	0.9.0
	- Dot product with conjugate - SNR computation - Euclidian length computation - Degrees to radians conversion - Radians to degrees conversion The cycle count estimate for DSP LAL is presented in the DSP LAL Cycle Count Estimate document.	1.0.0

Module	Features	Introduced in release
	-Added new functions for upper and lower triangular rectangular matrix inverse, real and complex single precision	0.8.3
	- Code optimization: Add restrict qualifier to output parameters that cannot be in-place memory with input parameters. Fixed intrinsic BBE_MAGN_2XCF32 inefficiency in functions	0.8.10
DSP Host Driver	Enables scheduling of processing jobs on the BBE32EP directly from the host CPU.	0.8.0 CD2
	Internal code refactoring	0.9.0
	Added support for ack messages between DSPHD and dispatcher Added command to check DSP dispatcher's responsiveness Added CRC check for every command IPCF integration for communication with DSP Dispatcher	1.0.0
	DSP-Host Driver available on S32R41 as AUTOSAR Complex Device Driver. Include Tresos plugin.	0.8.2
	CRC can now be configured (enabled/disabled) via API for DSP kernels Updated IPCF configuration to use new interrupt registration convention. Updated Complex Device Driver and plugin for AUTOSAR R21-11 Improved error reporting mechanism	0.8.4
	Added support to provide information when DSP enters IDLE state.	0.8.10
Radar Front-End API	New RFE Abstract 2.0 API implementation as AUTOSAR Complex Device Driver on M7 cores. Tresos plugin included. RFE Firmware for Single TEF82XX provided.	0.8.2
	Two TEF82XX cascaded mode support. TEF82XX calibration optimizations: - TX Power-out calibration - Calibration is performed only for profiles used in current sequence	0.8.3
	AUTOSAR R21-11 support Phase rotator support added Frequency autodrift	0.8.4
	TEF82XX Low-level driver 9.2.0 integration Functional Safety features: RF BIST, TX Integrity test, CAFC Integrity test, GPIO toggle test, Ball-break detection, Radar Front-end error reporting	0.8.5
	No new features added	0.8.6
	Obtain unique Hw ID from RFE to match T0 RFBIST data	0.8.7
	QPSK I/Q Line control using Dynamic Tables	0.8.8
	Temperature Monitor implementation	

Module	Features	Introduced in release
	Front-end registers dumping (CDD_Rfe_GetRegistryDump) Front-end monitoring (CDD_Rfe_MonitorRead) Front-end parameter updates (during radar cycle) : CDD_Rfe_UpdateBegin, CDD_Rfe_UpdateParam, CDD_Rfe_UpdatePush Phase rotators safety check (performed during radar cycle)	0.8.9
	TEF82XX Low-level driver 9.3.0 integration	0.8.10
CSI2 Driver	CSI2 Driver available on S32R41 as AUTOSAR Complex Device Driver. Include Tresos plugin.	0.8.2
	AUTOSAR R21-11 support	0.8.4
	S32DS Peripheral support	0.8.8
CTE Driver	CTE driver available on S32R41 as AUTOSAR Complex Device Driver. Include Tresos plugin.	0.8.2
	AUTOSAR R21-11 support	0.8.4
RFE with processing Example	Demonstrates the integration of the RSDK RFE Abstract 2.0 with data acquisition and processing components on the S32R41 EVB board, with or without a TEF82XX radar front-end connected to it.	0.8.2
	Integrated support for cascaded front-ends	0.8.3
	Added possibility to load RF configuration at runtime	
	Renamed project, changed structure, included M7_1 with RTD platform setup under same project	0.8.4
	Integrated RF BIST test in example app	0.8.5
	Obtain RFE FW Version and check against Blob data. Store/Retrieve/Validate T0 Data from file on host. Auto-trigger T0 Measurement for RF-BIST, in case of ID mismatch. Improved FUSA Fault reporting	0.8.7
	EB-Tresos plugin. S32DS Peripheral support.	0.8.8
	Introduced CDD_Rfe_GetRegisterDump function. Added ATB voltage read to this function.	0.8.10
	Introduced CRC for binary blob.	
RSDK Offline Example	Re-introduced the basic example showing data processing on SPT and BBE32 without a connected front-end	0.8.3
	Added S32 Debug Probe support.	0.8.4
DSP Dispatcher Example	Create an executable image for BBE32 which integrates the DSP Dispatcher and Radar BB Algos. The image can be loaded by another core. Designed to work together with other RSDK demo applications for A53.	0.8.0
	Added shared memory for communication between DSP Host Driver and DSP Dispatcher	0.8.0 CD2
	linked with optimized version of the XOS library	0.8.3
	Fixed MPU configuration example.	0.8.4
Common code	Basic application support tools: Status and Error reporting, File I/O using Lauterbach Trace32 Debugger	0.8.0

Module	Features	Introduced in release
Tools	Include RSDK Trace Logger and Parser: Lightweight, platform-independent trace logging mechanism for application-level debug and profiling on Zephyr.	0.8.0
	Added RSDK Trace logging support for DSP Dispatcher	0.9.0
	Extended the list of Matlab utility functions available	1.0.0
	Front End Config Generator: • Enables the configuration of all aspects of a Radar Cycle • Added support for generating multiple configurations (cascaded use-case) • Updated several parameter availabilities A complete list of historical changes is available in the PDF manual, available in /Tools/RFE2_Config/TEF82xx RFE Config Generator User Manual.pdf	0.8.3

3. Upgrading from the Previous Release

3.1 General Guidelines

This is a standalone package that needs to be installed separately from previous releases. **It is not compatible with previous releases for S32R41 using Zephyr OS (the latest one being Radar_SDK_1_0_0_RFP_230320).**

To update the RSDK components in an existing software application please check the following items:

- In your build and execution environment: use the tool versions described in section 1.1.
- Consult the tools and hardware installation instructions, as there may exist important notes, independent of this RSDK release.
- Update the existing projects to align the new AUTOSAR environment
- Use the EB-Tresos application and drivers plugins to setup the drivers usage.
- Use the latest RSDK API (described in RSDK User Manual)
 - update names of the libraries and API header files for drivers, accelerator kernels and common code
 - update names of changed API functions and data structures (check headers in /<Module>/api/ and /api for changes)
 - configure the new API fields and function calls

4. Usage of Third Party Software and Licensing

Radar SDK is intended to be used with other NXP software, as described in section 1.1.

Full licensing information is included in “SoftwareContentRegister_SWS32R41_Radar_SDK_0.8.10_D2412.txt” separate file.

The explicit license file for this release is LA_OPT_NXP_Software_License.txt.

Appendix A: Document History

VERSION	DATE	DESCRIPTION
0.1	December 2024	FIRST DRAFT FOR RELEASE 0.8.10
1.0	December 18 th 2024	RELEASED THE DOCUMENT

Appendix B: Package Release History

RELEASE VERSION	DATE	DESCRIPTION
0.8.0	Nov 26 2021	First RSDK release for S32R41 (bring-up)
0.8.0 CD2	11 Feb 2022	2 nd Code Drop
0.8.1	May 31 2022	Early Access release
0.9.0	Nov 25 2022	PRC Release
1.0.0	Mar 20 2023	RFP Release
0.8.2 CD	May 31 2023	Code Drop for introducing NXP RTOS and RFE Abstract 2.0
0.8.3 CD	July 31, 2023	Code Drop for adding support for cascaded TEF82XX front-ends.
0.8.4	October 24, 2023	Early Access Release
0.8.5	Dec 29 2023	Code Drop
0.8.6	Jan 17 2024	Code Drop

0.8.7	March 7 2024	Code Drop
0.8.8	April 12 2024	Code Drop
0.8.9	July 5 2024	Early Access Release
0.8.10	December 18 2024	Product Release Candidate

5. Legal information

5.1 Definitions

Draft — The document is a draft version only. The content is still under internal review and subject to formal approval, which may result in modifications or additions. NXP Semiconductors does not give any representations or warranties as to the accuracy or completeness of information included herein and shall have no liability for the consequences of use of such information.

5.2 Disclaimers

Limited warranty and liability — Information in this document is believed to be accurate and reliable. However, NXP Semiconductors does not give any representations or warranties, expressed or implied, as to the accuracy or completeness of such information and shall have no liability for the consequences of use of such information. NXP Semiconductors takes no responsibility for the content in this document if provided by an information source outside of NXP Semiconductors.

In no event shall NXP Semiconductors be liable for any indirect, incidental, punitive, special or consequential damages (including - without limitation - lost profits, lost savings, business interruption, costs related to the removal or replacement of any products or rework charges) whether or not such damages are based on tort (including negligence), warranty, breach of contract or any other legal theory.

Notwithstanding any damages that customer might incur for any reason whatsoever, NXP Semiconductors' aggregate and cumulative liability towards customer for the products described herein shall be limited in accordance with the *Terms and conditions of commercial sale* of NXP Semiconductors.

Right to make changes — NXP Semiconductors reserves the right to make changes to information published in this document, including without limitation specifications and product descriptions, at any time and without notice. This document supersedes and replaces all information supplied prior to the publication hereof.

Suitability for use — NXP Semiconductors products are not designed, authorized or warranted to be suitable for use in life support, life-critical or safety-critical systems or equipment, nor in applications where failure or malfunction of an NXP Semiconductors product can reasonably be expected to result in personal injury, death or severe property or environmental damage. NXP Semiconductors and its suppliers accept no liability for inclusion and/or use of NXP Semiconductors products in such equipment or applications and therefore such inclusion and/or use is at the customer's own risk.

Applications — Applications that are described herein for any of these products are for illustrative purposes only. NXP Semiconductors makes no representation or warranty that such applications will be suitable for the specified use without further testing or modification.

Customers are responsible for the design and operation of their applications and products using NXP Semiconductors products, and NXP Semiconductors accepts no liability for any assistance with applications or customer product design. It is customer's sole responsibility to determine whether the NXP Semiconductors product is suitable and fit for the customer's applications and products planned, as well as for the planned application and use of customer's third party customer(s). Customers should provide appropriate design and operating safeguards to minimize the risks associated with their applications and products.

NXP Semiconductors does not accept any liability related to any default, damage, costs or problem which is based on any weakness or default in the customer's applications or products, or the application or use by customer's third party customer(s). Customer is responsible for doing all necessary testing for the customer's applications and products using NXP Semiconductors products in order to avoid a default of the applications and the products or of the application or use by customer's third party customer(s). NXP does not accept any liability in this respect.

Export control — This document as well as the item(s) described herein may be subject to export control regulations. Export might require a prior authorization from competent authorities.

Translations — A non-English (translated) version of a document is for reference only. The English version shall prevail in case of any discrepancy between the translated and English versions.

Evaluation products — This product is provided on an "as is" and "with all faults" basis for evaluation purposes only. NXP Semiconductors, its affiliates and their suppliers expressly disclaim all warranties, whether express, implied or statutory, including but not limited to the implied warranties of noninfringement, merchantability and fitness for a particular purpose. The entire risk as to the quality, or arising out of the use or performance, of this product remains with customer.

In no event shall NXP Semiconductors, its affiliates or their suppliers be liable to customer for any special, indirect, consequential, punitive or incidental damages (including without limitation damages for loss of business, business interruption, loss of use, loss of data or information, and the like) arising out of the use of or inability to use the product, whether or not based on tort (including negligence), strict liability, breach of contract, breach of warranty or any other theory, even if advised of the possibility of such damages.

Notwithstanding any damages that customer might incur for any reason whatsoever (including without limitation, all damages referenced above and all direct or general damages), the entire liability of NXP Semiconductors, its affiliates and their suppliers and customer's exclusive remedy for all of the foregoing shall be limited to actual damages incurred by customer based on reasonable reliance up to the greater of the amount actually paid by customer for the product or five dollars (US\$5.00). The foregoing limitations, exclusions and disclaimers shall apply to the maximum extent permitted by applicable law, even if any remedy fails of its essential purpose.

All information provided in this document is subject to legal disclaimers.

© NXP B.V. 2024. All rights reserved.

Please be aware that important notices concerning this document and the product(s) described herein, have been included in the section 'Legal information'.

© NXP B.V. 2024.

All rights reserved.

For more information, please visit: <http://www.nxp.com>
For sales office addresses, please send an email to: salesaddresses@nxp.com

Date of release: December 18th2024